Zongyi Li

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RESEARCH INTEREST

My research interests lie at the intersection of machine learning and physical sciences (AI for science). Specifically, I work on neural operators for learning solution operators in partial differential equations (PDEs) that arise in fluid mechanics and earth sciences.

APPOINTMENTS

Assistant Professor, NYU Courant 2026 - present
Mathematics and Data Science. Courant Institute of Mathematical Sciences
Center for Atmosphere Ocean Science (CAOS) and Center for Data Science (CDS)

Postdoc, MIT 2025 - 2026
Computer Science and Artificial Intelligence Laboratory (CSAIL). Hosted by Kaiming He

EDUCATION

Ph.D., Caltech 2019 - 2025
Computing and Mathematical Science. Advised by Anima Anandkumar
Milton and Francis Clauser Doctoral Prize (best thesis award)

B.S., Washington University in St. Louis 2015 - 2019
Mathematics and Computer Science
Ross Middlemiss award in mathematics and the highest distinction

WORK EXPERIENCE

Research Internships, Nvidia Summer 2022, 2023, 2024
Worked on developing machine learning models for scientific applications

AWARDS

Schmidt Science AI2050 Early Career Fellowship 2026
NVIDIA Academic Grant Program 2026
MIT-Novo Nordisk AI Postdoctoral Fellowship 2025
Milton and Francis Clauser Doctoral Prize 2025
(Awarded to a single Caltech Ph.D. graduate whose thesis exhibits the highest originality)
Caltech Graduate Teaching and Mentoring Award 2025
Jane Street Fellowship - Finalist 2024
Nvidia Fellowship 2023
Amazon AI4Science Fellowship 2022
PIMCO Fellowship 2021
Kortschak Scholars Fellowship 2019

PUBLICATIONS

[1] **Li, Zongyi***, Jasmine Shone*, Harry Gong, Luoxi Wu, Yehong Jiang, Michael Yang, Zander Li, Minghao Guo, and Kaiming He. "Generative Neural Operator as a Physics Foundation Model". Preprint, in submission. 2026.

[2] Harry Gong*, Yehong Jiang*, Kaiming He, and **Li, Zongyi**. "A Distributional Evaluation Metric for Chaotic Physical Systems". Preprint, in submission. 2026. *Equal contribution.

[3] Jean Kossaifi, Nikola Kovachki, **Li, Zongyi**, David Pitt, Miguel Liu-Schiaffini, Robert Joseph George, Boris Bonev, Kamyar Azizzadenesheli, Julius Berner, Valentin Duruisseaux, et al. "A Library for Learning Neural Operators". In: *Journal of Machine Learning Research* 27.205 (2026), pp. 1–6.

[4] Jingyi Qiu, Hong Chen, and **Li, Zongyi**. "Counterfactual LLM-based Framework for Measuring Rhetorical Style". In: *International Conference on Learning Representations*. 2026.

[5] Haixu Wu, Minghao Guo, **Li, Zongyi**, Zhiyang Dou, Mingsheng Long, Kaiming He, and Wojciech Matusik. "GeoPT: Scaling Physics Simulation via Lifted Geometric Pre-Training". In: *International Conference on Machine Learning (ICML)*. 2026.

[6] **Li, Zongyi**, Samuel Lanthaler, Catherine Deng, Michael Chen, Yixuan Wang, Kamyar Azizzadenesheli, and Anima Anandkumar. "Scale-Consistent Learning for Partial Differential Equations". arXiv preprint arXiv:2507.18813. 2025.

[7] Kamyar Azizzadenesheli, Nikola Kovachki, **Li, Zongyi**, Miguel Liu-Schiaffini, Jean Kossaifi, and Anima Anandkumar. "Neural operators for accelerating scientific simulations and design". In: *Nature Reviews Physics* (2024), pp. 1–9.

[8] Robert Joseph George, Jiawei Zhao, Jean Kossaifi, **Li, Zongyi**, and Anima Anandkumar. "Incremental spatial and spectral learning of neural operators for solving large-scale PDEs". In: *Transactions on Machine Learning Research (TMLR)* (2024).

[9] Vignesh Gopakumar, Stanislas Pamela, Lorenzo Zanisi, **Li, Zongyi**, Anima Anandkumar, and MAST Team. "Fourier neural operator for plasma modelling". In: *Nuclear Fusion* (2024).

[10] Samuel Lanthaler, **Li, Zongyi**, and Andrew M Stuart. "The nonlocal neural operator: Universal approximation". In: *Constructive Approximation* (2024).

[11] **Li, Zongyi**, Hongkai Zheng, Nikola Kovachki, David Jin, Haoxuan Chen, Burigede Liu, Kamyar Azizzadenesheli, and Anima Anandkumar. "Physics-informed neural operator for learning partial differential equations". In: *ACM/JMS Journal of Data Science* 1.3 (2024), pp. 1–27.

[12] Md Ashiqur Rahman, Robert Joseph George, Mogab Elleithy, Daniel Leibovici, **Li, Zongyi**, Boris Bonev, Colin White, Julius Berner, Raymond A Yeh, Jean Kossaifi, et al. "Pretraining codomain attention neural operators for solving multiphysics PDEs". In: *Advances in Neural Information Processing Systems (NeurIPS)*. 2024.

[13] Chuwei Wang, Julius Berner, **Li, Zongyi**, Di Zhou, Jiayun Wang, Jane Bae, and Anima Anandkumar. "Beyond Closure Models: Learning Chaotic-Systems via Physics-Informed Neural Operators". arXiv preprint arXiv:2408.05177. 2024.

[14] Tingtao Zhou, Xuan Wan, Daniel Zhengyu Huang, **Li, Zongyi**, Zhiwei Peng, Anima Anandkumar, John F Brady, Paul W Sternberg, and Chiara Daraio. "AI-aided geometric design of anti-infection catheters". In: *Science Advances* 10.1 (2024), eadj1741.

[15] Nikola Kovachki, **Li, Zongyi**, Burigede Liu, Kamyar Azizzadenesheli, Kaushik Bhattacharya, Andrew Stuart, and Anima Anandkumar. "Neural operator: Learning maps between function spaces with applications to PDEs". In: *Journal of Machine Learning Research* 24.89 (2023), pp. 1–97.

[16] **Li, Zongyi**, Daniel Zhengyu Huang, Burigede Liu, and Anima Anandkumar. "Fourier neural operator with learned deformations for PDEs on general geometries". In: *Journal of Machine Learning Research* 24.388 (2023), pp. 1–26.

[17] **Li, Zongyi**, Nikola Kovachki, Chris Choy, Boyi Li, Jean Kossaifi, Shourya Otta, Mohammad Amin Nabian, Maximilian Stadler, Christian Hundt, Kamyar Azizzadenesheli, et al. "Geometry-informed neural operator for large-scale 3D PDEs". In: *Advances in Neural Information Processing Systems (NeurIPS)* 36 (2023).

[18] Jaideep Pathak, Shashank Subramanian, Peter Harrington, Sanjeev Raja, Ashesh Chattopadhyay, Morteza Mardani, Thorsten Kurth, David Hall, **Li, Zongyi**, Kamyar Azizzadenesheli, et al. "FourCastNet: A global data-driven high-resolution weather model using adaptive Fourier neural operators". In: *PASC '23: Proceedings of the Platform for Advanced Scientific Computing Conference*. 2023.

[19] Gege Wen, **Li, Zongyi**, Qirui Long, Kamyar Azizzadenesheli, Anima Anandkumar, and Sally M Benson. "Real-time high-resolution CO_2 geological storage prediction using nested Fourier neural operators". In: *Energy & Environmental Science* 16.4 (2023), pp. 1732–1741.

[20] Colin White, Julius Berner, Jean Kossaifi, Mogab Elleithy, David Pitt, Daniel Leibovici, **Li, Zongyi**, Kamyar Azizzadenesheli, and Anima Anandkumar. "Physics-informed neural operators with exact differentiation on arbitrary geometries". In: *The Symbiosis of Deep Learning and Differential Equations III*. 2023.

[21] John Guibas, Morteza Mardani, **Li, Zongyi**, Andrew Tao, Anima Anandkumar, and Bryan Catanzaro. "Efficient Token Mixing for Transformers via Adaptive Fourier Neural Operators". In: *International Conference on Learning Representations*. 2022.

[22] **Li, Zongyi**, Miguel Liu-Schiaffini, Nikola Kovachki, Kamyar Azizzadenesheli, Burigede Liu, Kaushik Bhattacharya, Andrew Stuart, and Anima Anandkumar. "Learning chaotic dynamics in dissipative systems". In: *Advances in Neural Information Processing Systems (NeurIPS)*. 2022.

[23] Yuanyuan Shi, **Li, Zongyi**, Huan Yu, Drew Steeves, Anima Anandkumar, and Miroslav Krstic. "Machine learning accelerated PDE backstepping observers". In: *2022 IEEE 61st Conference on Decision and Control (CDC)*. IEEE. 2022, pp. 5423–5428.

[24] Anda Trifan, Defne Gorgun, **Li, Zongyi**, Alexander Brace, Maxim Zvyagin, Heng Ma, Austin R Clyde, David A Clark, Michael Salim, David Hardy, et al. "Intelligent Resolution: Integrating Cryo-EM with AI-driven Multi-resolution Simulations to Observe the SARS-CoV-2 Replication-Transcription Machinery in Action". In: *The International Journal of High Performance Computing Applications* (2022).

[25] Gege Wen, **Li, Zongyi**, Kamyar Azizzadenesheli, Anima Anandkumar, and Sally M Benson. "U-FNO—An enhanced Fourier neural operator-based deep-learning model for multiphase flow". In: *Advances in Water Resources* 163 (2022), p. 104180.

[26] Haoyu Yang, **Li, Zongyi**, Kumara Sastry, Saumyadip Mukhopadhyay, Mark Kilgard, Anima Anandkumar, Brucek Khailany, Vivek Singh, and Haoxing Ren. "Generic lithography modeling with dual-band optics-inspired neural networks". In: *Proceedings of the 59th ACM/IEEE Design Automation Conference*. 2022, pp. 973–978.

[27] **Li, Zongyi**, Nikola Kovachki, Kamyar Azizzadenesheli, Burigede Liu, Kaushik Bhattacharya, Andrew Stuart, and Anima Anandkumar. "Fourier neural operator for parametric partial differential equations". In: *International Conference on Learning Representations*. 2021.

[28] Burigede Liu, Nikola Kovachki, **Li, Zongyi**, Kamyar Azizzadenesheli, Anima Anandkumar, Andrew Stuart, and Kaushik Bhattacharya. "A learning-based multiscale method and its application to inelastic impact problems". In: *Journal of the Mechanics and Physics of Solids* (2021).

[29] Diego Calderon, Brendan Juba, Sirui Li, **Li, Zongyi**, and Lisa Ruan. "Conditional Linear Regression". In: *International Conference on Artificial Intelligence and Statistics (AISTATS)*. 2020.

[30] **Li, Zongyi**, Nikola Kovachki, Kamyar Azizzadenesheli, Burigede Liu, Kaushik Bhattacharya, Andrew Stuart, and Anima Anandkumar. "Neural Operator: Graph Kernel Network for Partial Differential Equations". arXiv preprint arXiv:2003.03485. 2020.

[31] **Li, Zongyi**, Nikola Kovachki, Kamyar Azizzadenesheli, Burigede Liu, Andrew Stuart, Kaushik Bhattacharya, and Anima Anandkumar. "Multipole graph neural operator for parametric partial differential equations". In: *Advances in Neural Information Processing Systems (NeurIPS)* 33 (2020), pp. 6755–6766.

[32] Brendan Juba, **Li, Zongyi**, and Evan Miller. "Learning Abduction Using Partial Observability". In: *Proceedings of the AAAI Conference on Artificial Intelligence*. Vol. 32. 1. 2018.

SOFTWARE

- Neural Operator Library (founder, 3.4k stars) `https://github.com/neuraloperator`
- Modulus Library `https://github.com/NVIDIA/modulus`

- Clima Library `https://github.com/CliMA/TurbulenceConvection.jl`
- Tensorly Library `https://github.com/tensorly/tensorly`

MEDIA COVERAGE

- MIT Tech Review: AI has cracked a key mathematical puzzle for understanding our world.
- Quanta Magazine: Latest Neural Nets Solve World's Hardest Equations Faster Than Ever Before.
- Quanta Magazine: The Year in Math and Computer Science.
- Towards Data Science: AI has unlocked a key scientific hurdle in predicting our world.
- Medium: Artificial Intelligence Can Now Solve Partial Differential Equations.

INVITED TALKS

- IAIFI Summer Workshop, MIT: *Generative neural operator as a physics foundation model* Aug 2026
- Microsoft Research New England, Generative Modeling & Sampling Seminar: *Generative model for scientific computing* Sep 2025
- UCLA, hosted by Yizhou Sun and Wei Wang: *Neural operator for scientific computing* Nov 2024
- Rising Stars in Data Science Workshop, UCSD: *Scale-consistency in operator learning* Nov 2024
- UChicago, hosted by Pedram Hassanzadeh: *Neural operator for scientific computing* Oct 2024
- APS Division of Plasma Physics Meeting: *Scale-consistency in operator learning* Oct 2024
- UMichigan SciFM Summer School: *Scale-consistency in operator learning* July 2024
- NVIDIA GTC (Graduate Fellowship recipient talk): *Automotive and aerodynamic design using machine learning* March 2024
- CVPR Tutorial on Neural Fields: *Deformed spectral methods for general geometries* June 2022
- UCSD, guest lecture in Machine Learning for Physical Science (Yuanyuan Shi): *Neural operator for scientific computing* April 2022
- Caltech, guest lecture in Representation Learning for Science (Yisong Yue): *Neural operator for scientific computing* April 2022
- CMU, NSF AI Planning Institute for Data Discovery in Physics: *Physics-informed neural operator* Sep 2021
- CMU, ML in Fluid Dynamics series (DARPA-E and CMU SciML webinar): *Physics-informed neural operator* June 2021

SERVICES

Area Chair: ICLR

Reviewer:

- Machine Learning: Neurips, ICLR, ICML, AAAI, TPAMI, JMLR
- Computational Physics: COMMSPHYS, JCP, CMAME, APS-PRR, APS-PRF
- Applied Math: SIAM-JUQ, SIAM-SISC
- Geo-Physics: JGR-ML

TEACHING

Instructor at New York University

- MATH-UY 4114: Applied Statistics
  Fall 2026

Teaching Assistant at California Institute of Technology

- CS 165: Foundations of Machine Learning and Statistical Inference
  Winter 2021 (Head TA), Winter 2022, Winter 2023 (Head TA), Winter 2024

MENTORING

PhD students at New York University

- Jialin Chen (co-advised with Sara Shamekh)
- Hong Chul Nam
- Zhikai Wu

Undergraduate researchers at MIT, UROP program

- Jasmine Shone (2025-2026)
- Harry Gong (2025-2026)
- Yehong Jiang (2025-2026)
- Luoxi Wu (2025-2026)
- Michael Yang (2025-2026)
- Zander Li (2025-2026)

Undergraduate researchers at Caltech, SURF program

- David Jin (2021 $\rightarrow$ MIT PhD)
- Derek Qin (2021 $\rightarrow$ Databricks)
- Miguel Liu-Schiaffini (2021-2023 $\rightarrow$ Stanford PhD)
- Kimia Hassibi (2022 $\rightarrow$ MIT PhD)
- Haydn Maust (2022)
- Zelin Zhao (2023 $\rightarrow$ Gatech PhD)
- Catherine Deng (2023-2024 $\rightarrow$ Stanford PhD)
- Vansh Tibrewal (2023-2024)
- Xinyi Li (2024 $\rightarrow$ Caltech PhD)
- Reva Dhillon (2024)
- Michael Chen (2025)